

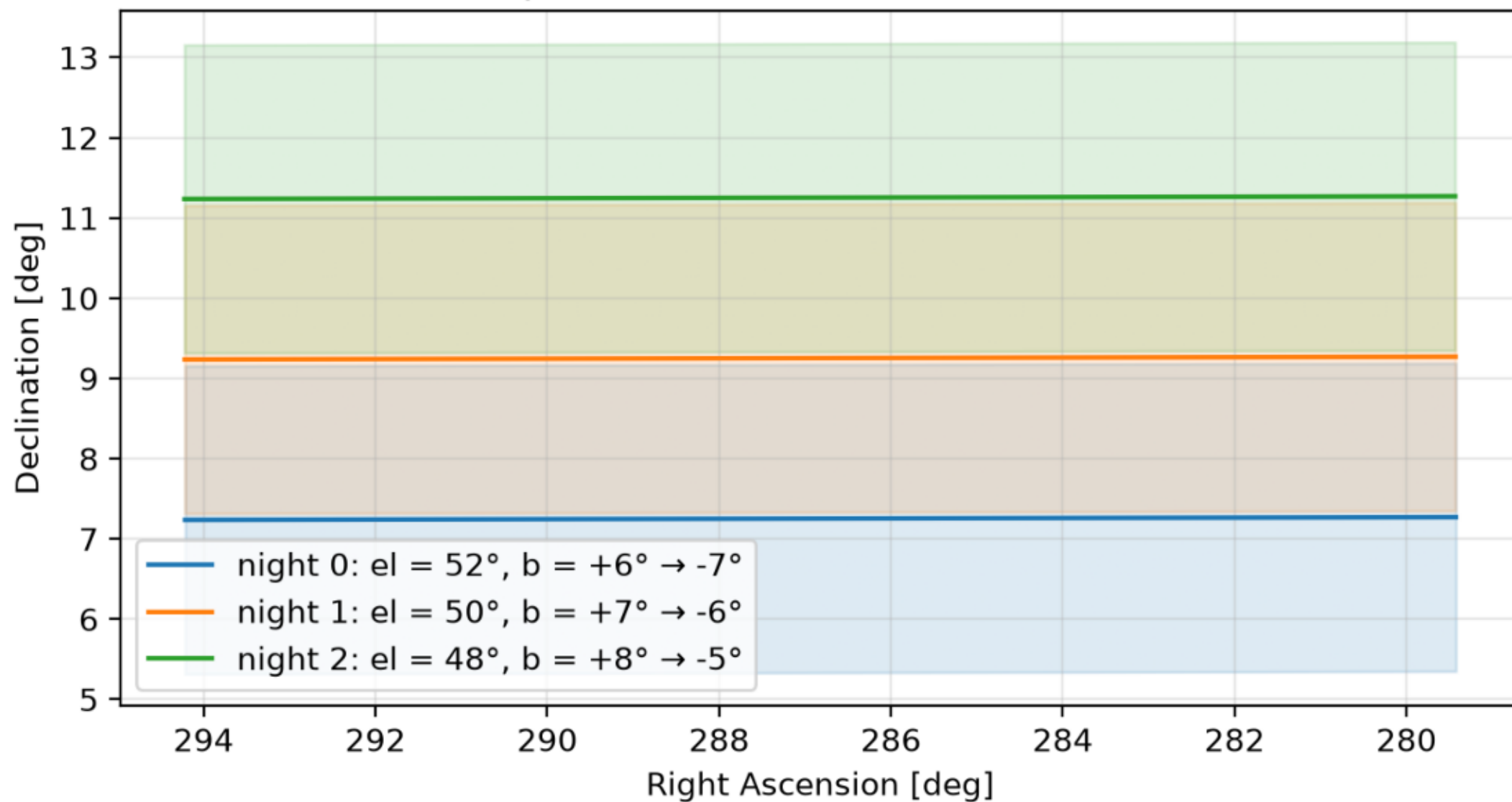
Experiment 003 — galplane-drift-sidelobe-stress

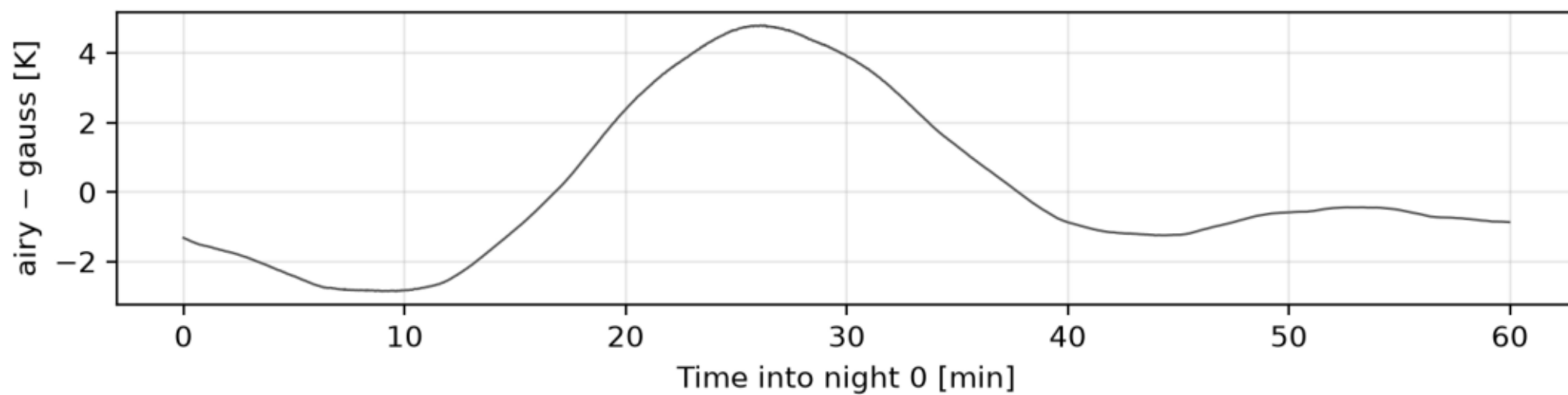
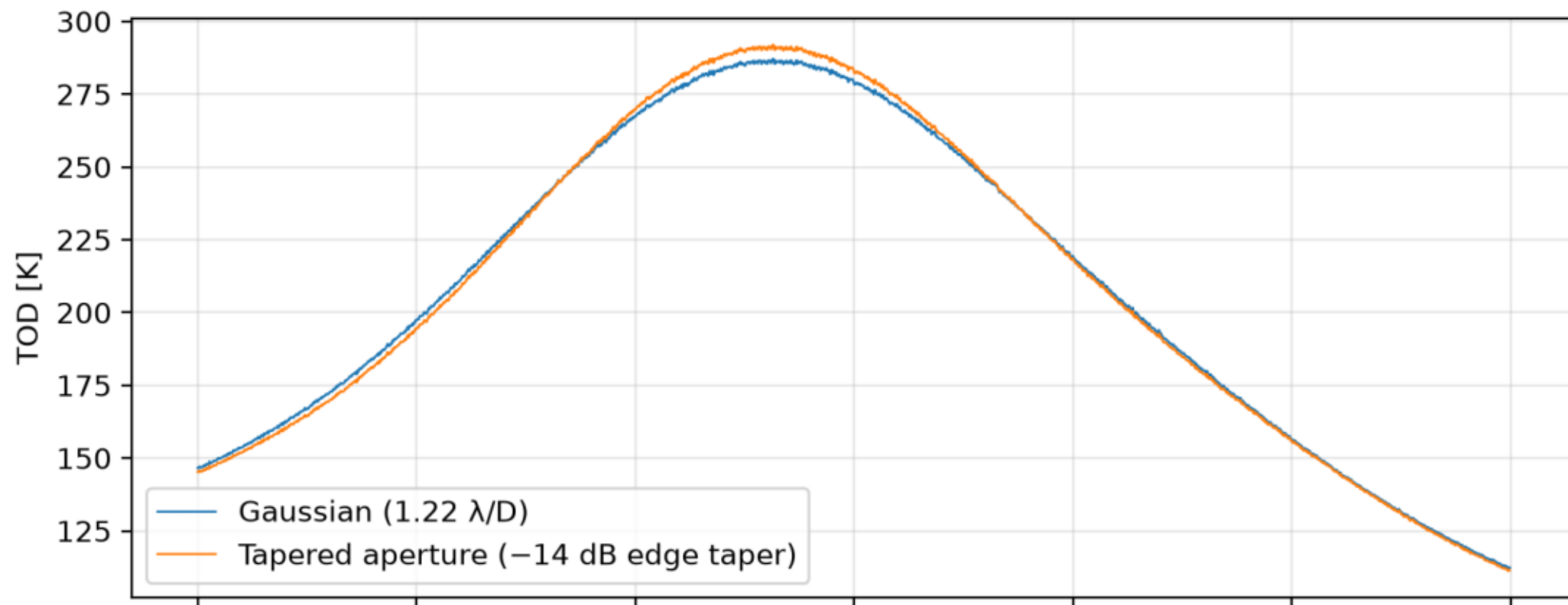
2026-07-28

Galactic-plane sidelobe stress test: identical drift design, beams, noise seeds and map-making recipe as experiments 001/002, but the LST window (start 2024-04-16 03:34:00 UTC, LST $\approx 280^\circ$) sweeps the Dec $\approx +9^\circ$ strip across the galactic plane ($b = +7^\circ \rightarrow -6^\circ$, crossing at $l \approx 43^\circ$). Scenarios: matched Gaussian, matched tapered aperture, and beam mismatch (tapered-aperture sky through the Gaussian operator). The mismatch — gauss rows isolate the unmodelled-sidelobe map error (same operator, mask, prior and noise realization). A 1/f noise budget re-solves the map with the flicker and white components switched off one at a time: the residual is dominated by the noiseless reconstruction floor (21.4 K), not by 1/f (4.8 K), and the 2 mHz high-pass is a net regression in this regime.

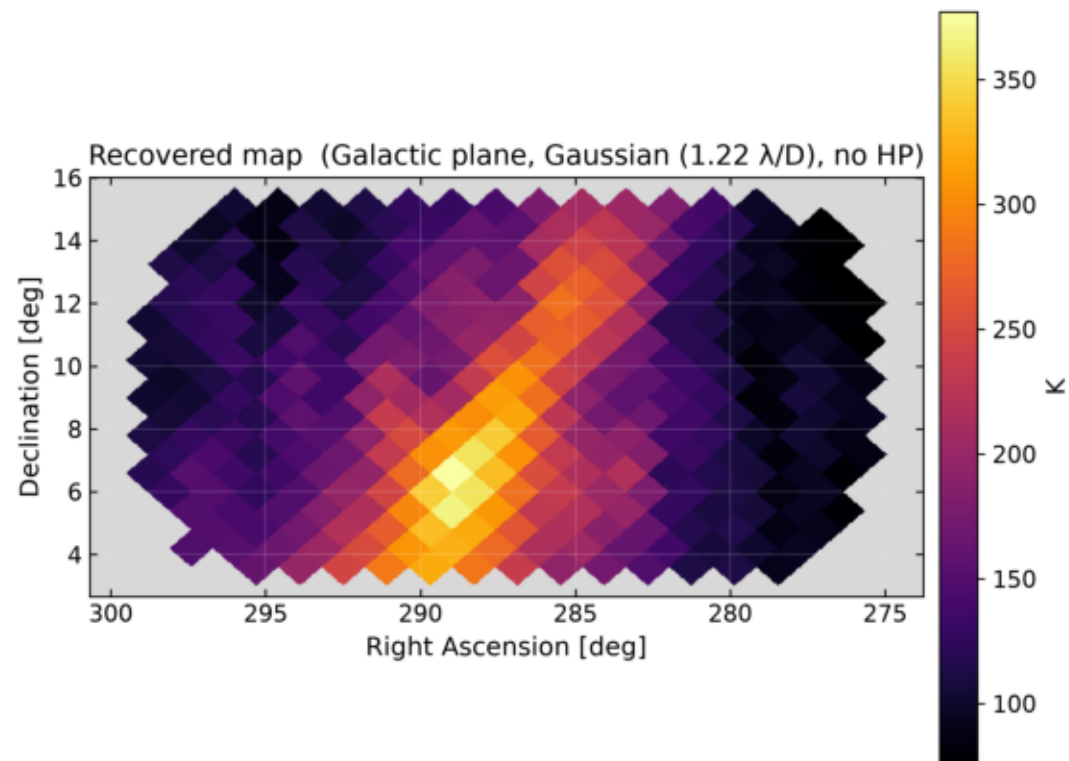
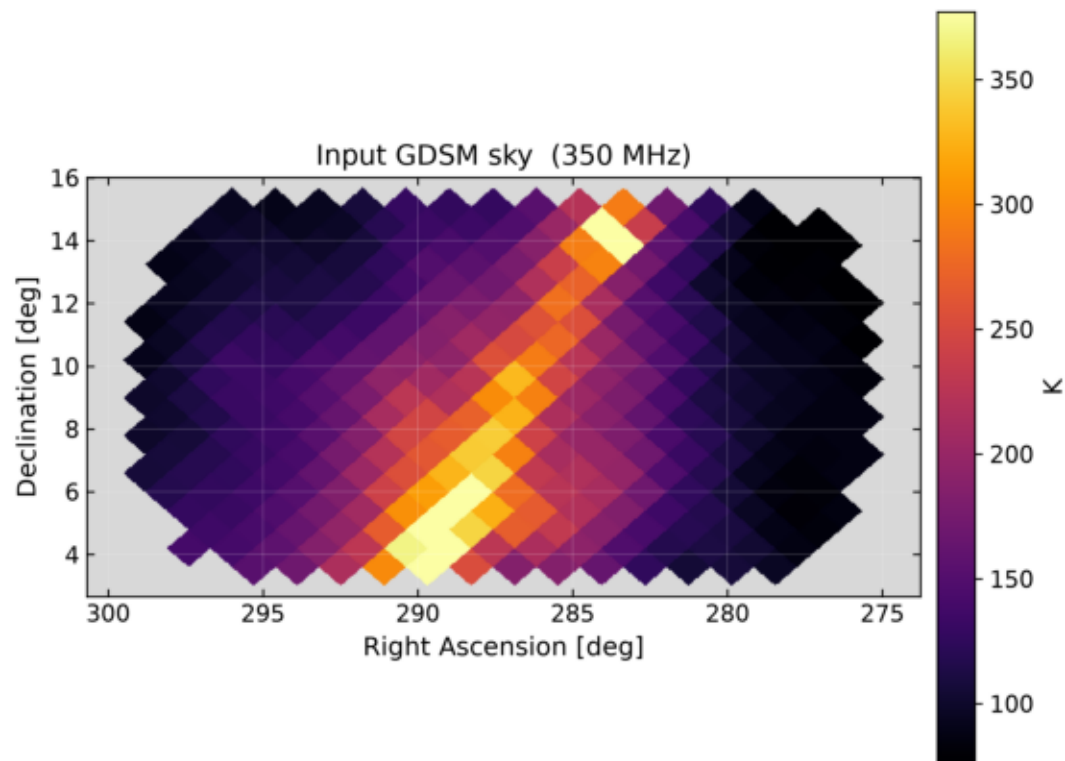
gauss / no HP	23081.4 mK
gauss / HP 2 mHz	31439.3 mK
airy / no HP	25262.0 mK
airy / HP 2 mHz	32206.9 mK
mismatch / no HP	23901.2 mK
mismatch / HP 2 mHz	30668.1 mK
mismatch - gauss / no HP	7975.3 mK
mismatch - gauss / HP 2 mHz	4496.9 mK
--- 1/f budget ---	
galactic plane / no HP: 1/f-induced	4770.1 mK
galactic plane / no HP: recon floor	21368.0 mK
galactic plane / HP 2 mHz: 1/f-induced	1869.1 mK
galactic plane / HP 2 mHz: recon floor	23426.0 mK
b = +50 deg / no HP: 1/f-induced	138.1 mK
b = +50 deg / no HP: recon floor	1287.0 mK
b = +50 deg / HP 2 mHz: 1/f-induced	35.3 mK
b = +50 deg / HP 2 mHz: recon floor	1107.2 mK

Galactic-plane drift tracks (bands = beam FWHM)

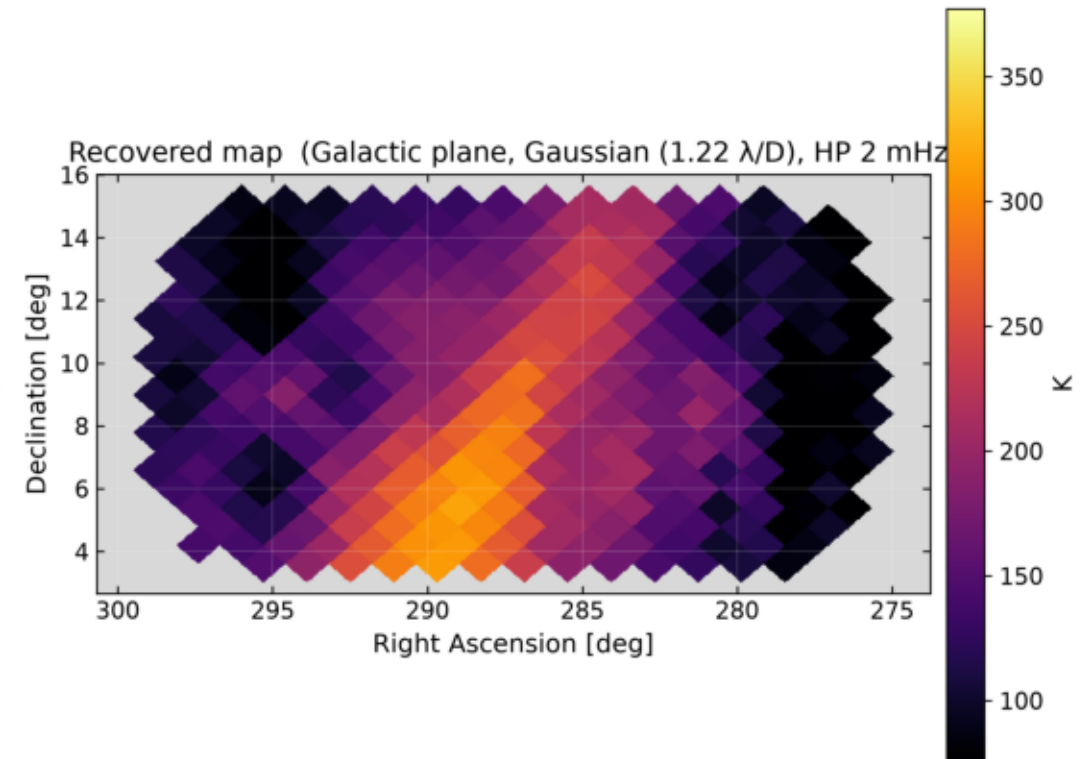
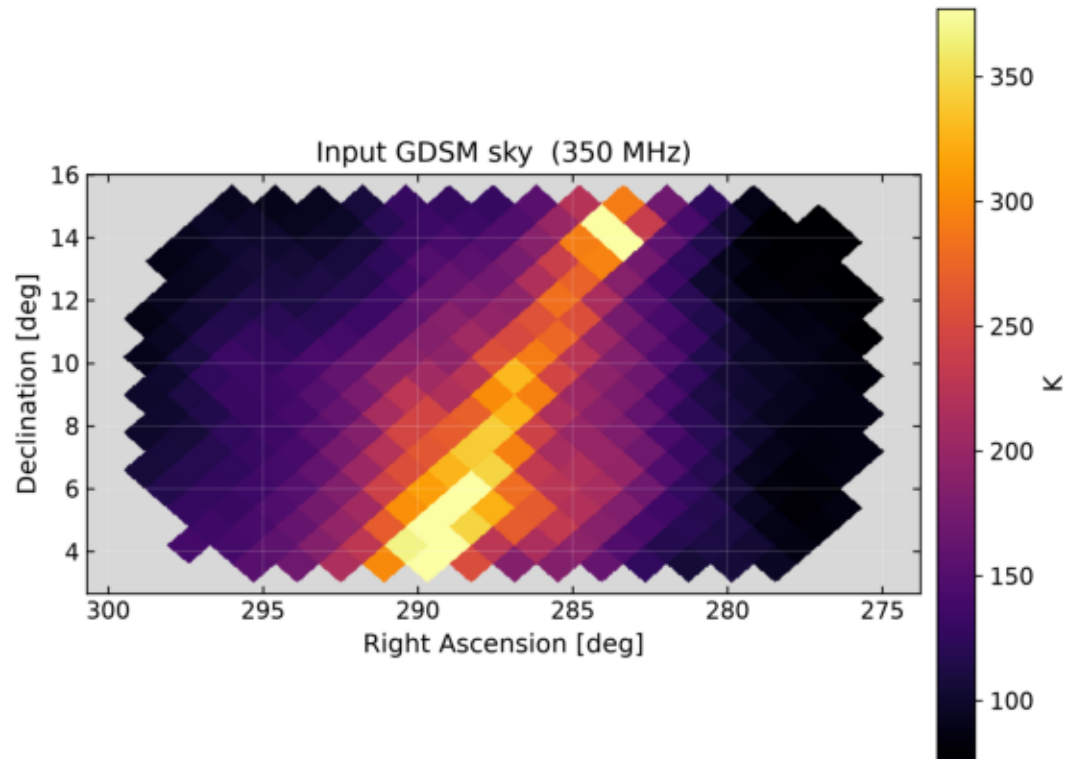




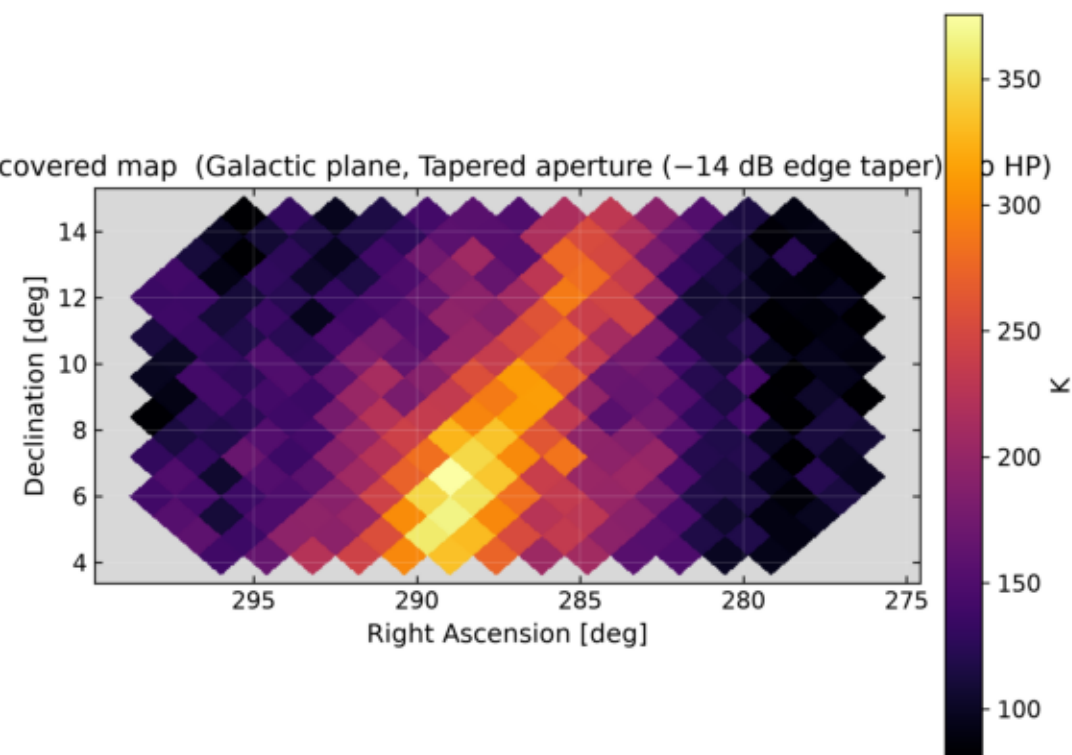
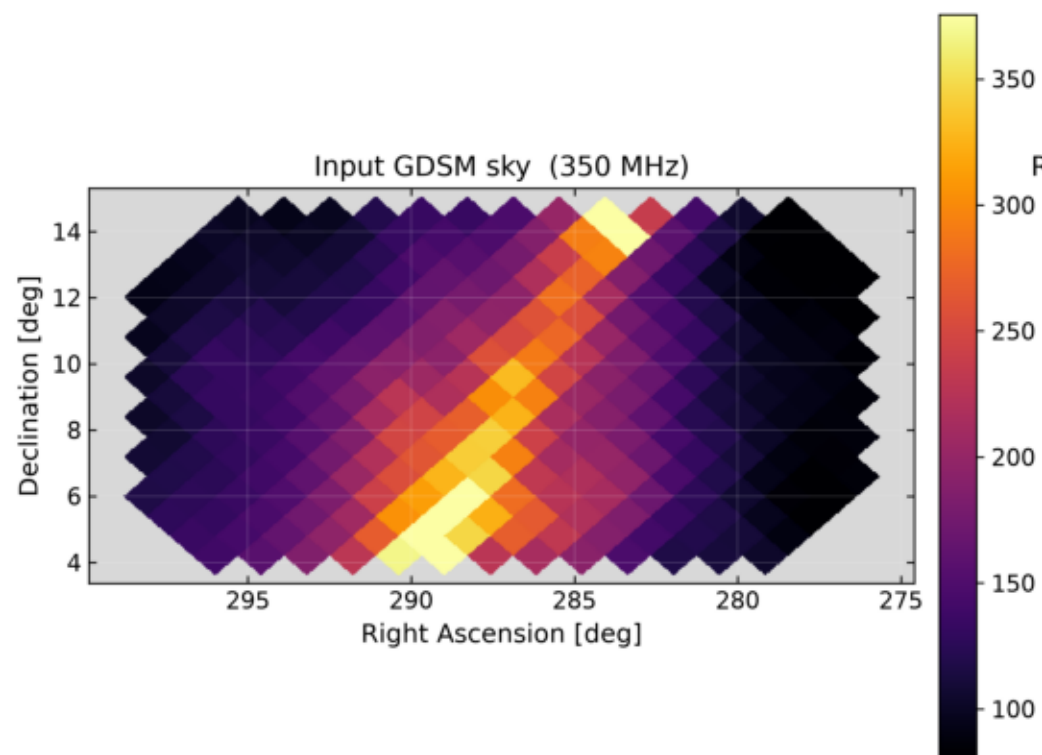
map_ska_galplane_gauss_noHP.png



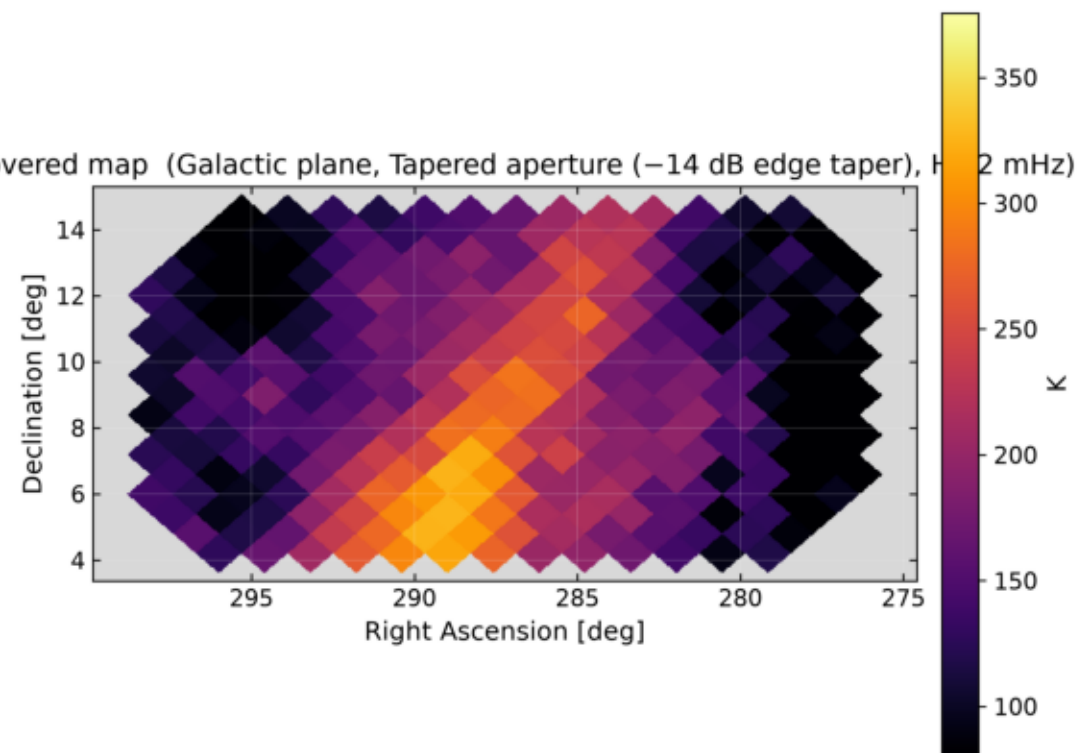
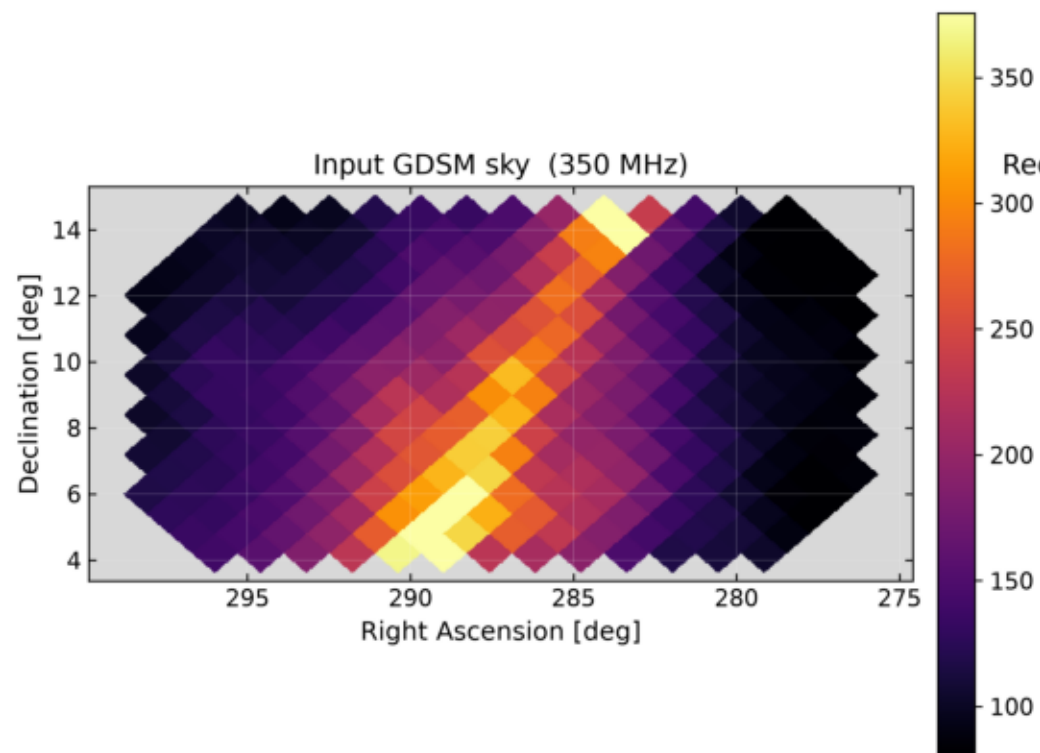
map_ska_galplane_gauss_HP.png



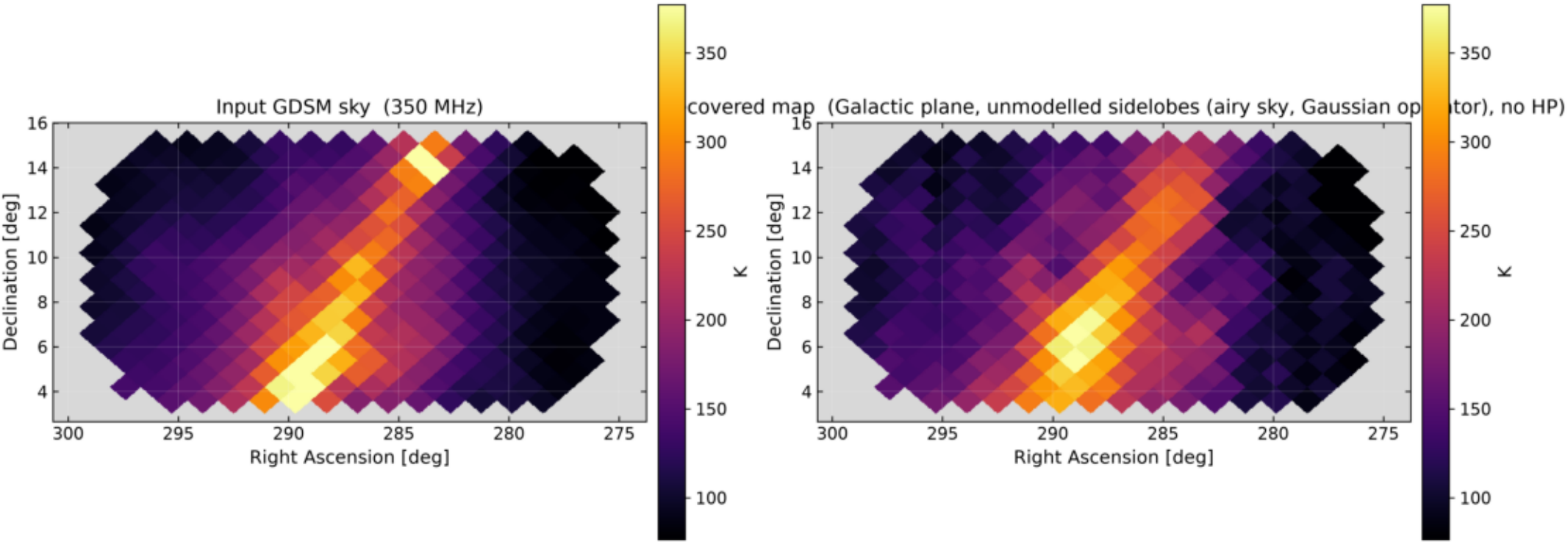
map_ska_galplane_airy_noHP.png



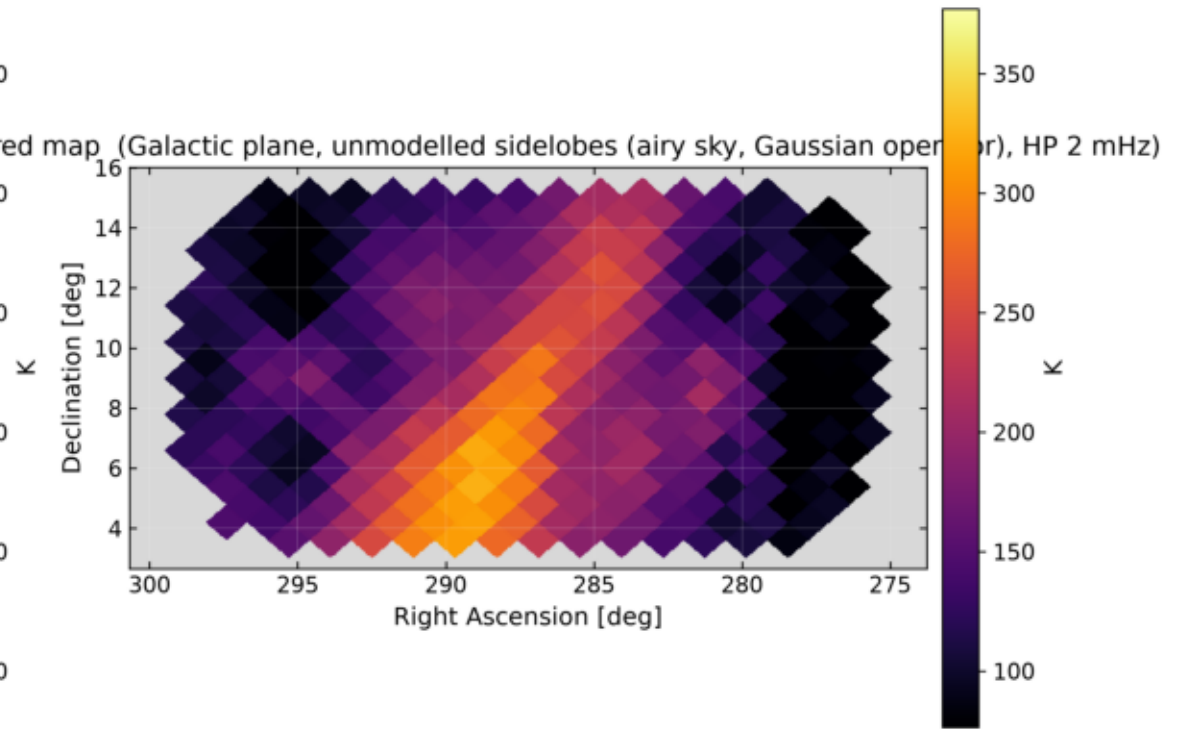
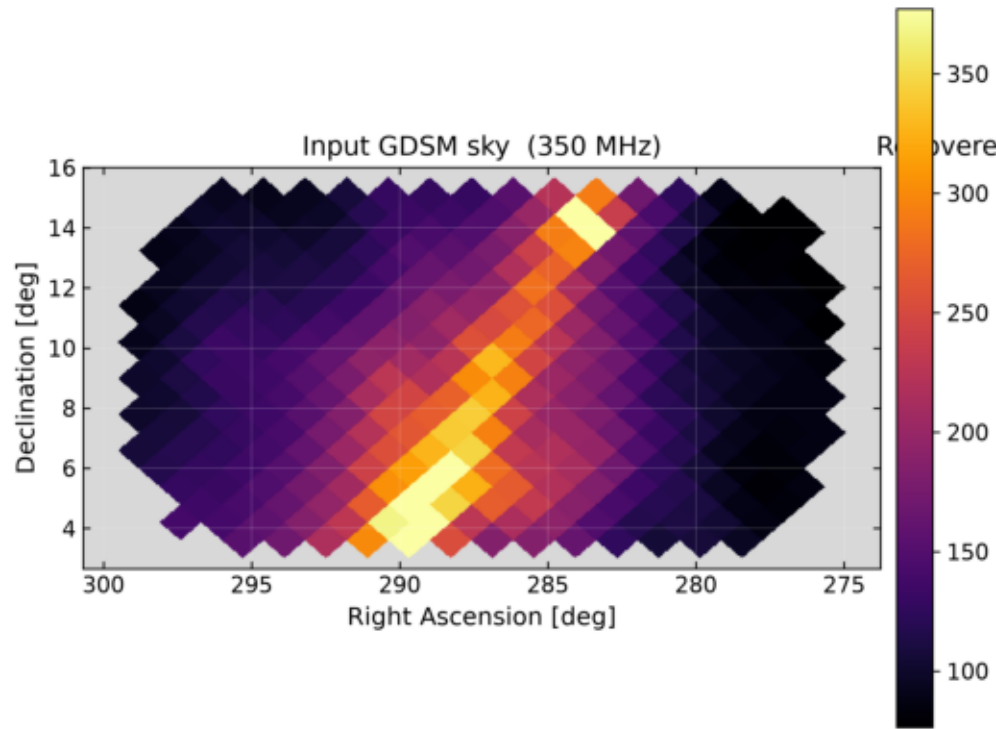
map_ska_galplane_airy_HP.png

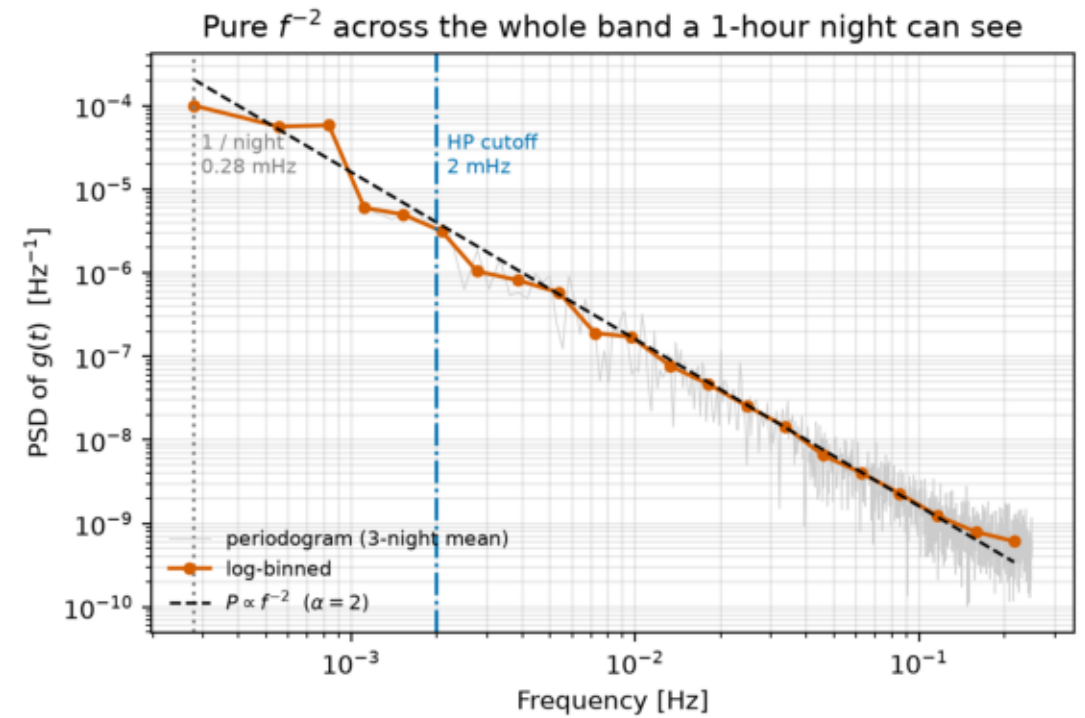
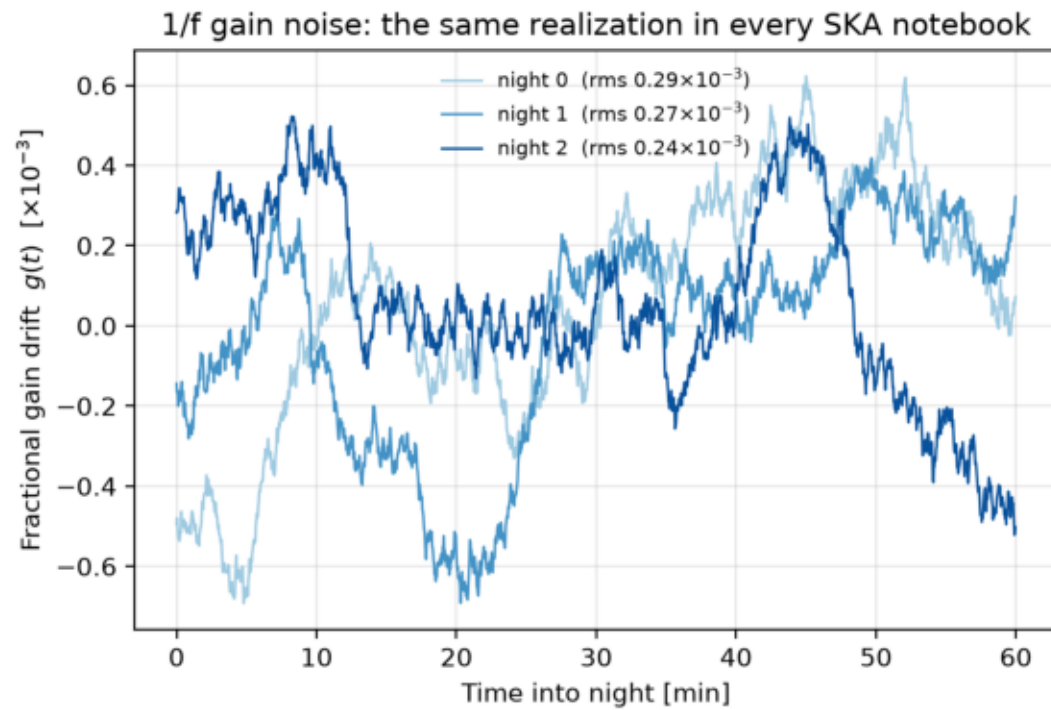


map_ska_galplane_mismatch_noHP.png

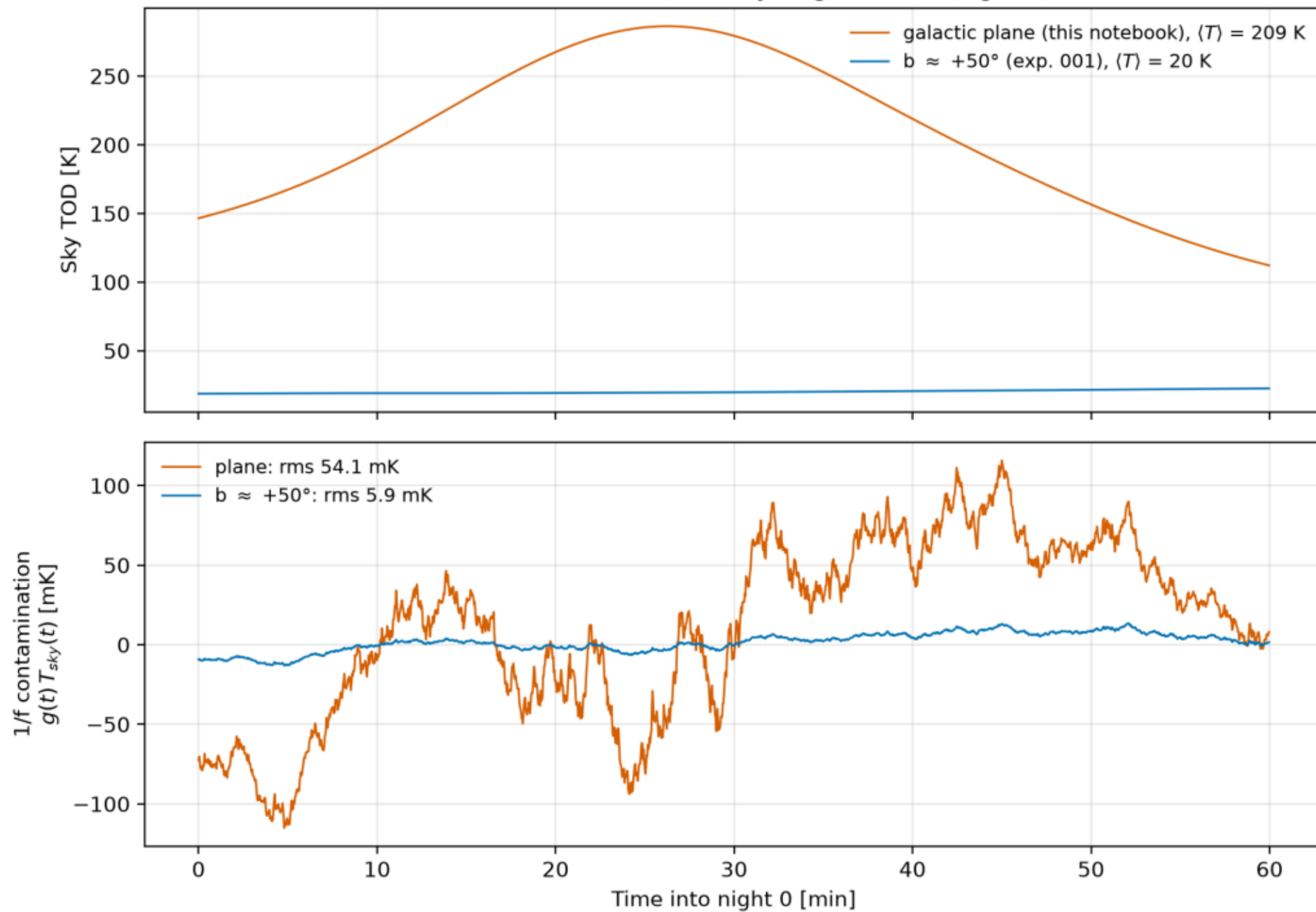


map_ska_galplane_mismatch_HP.png

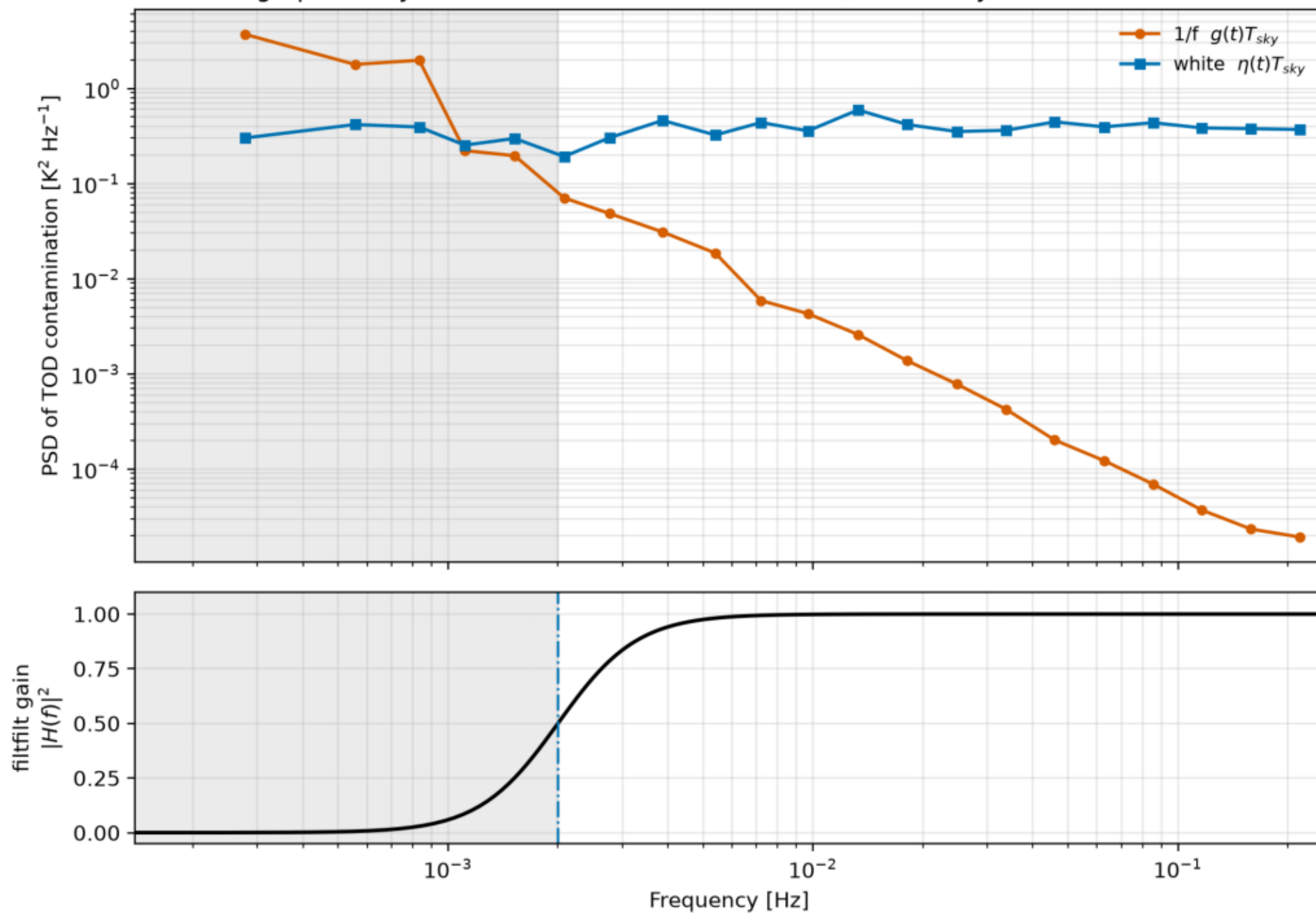




Same 1/f realization, two sky brightnesses (night 0)

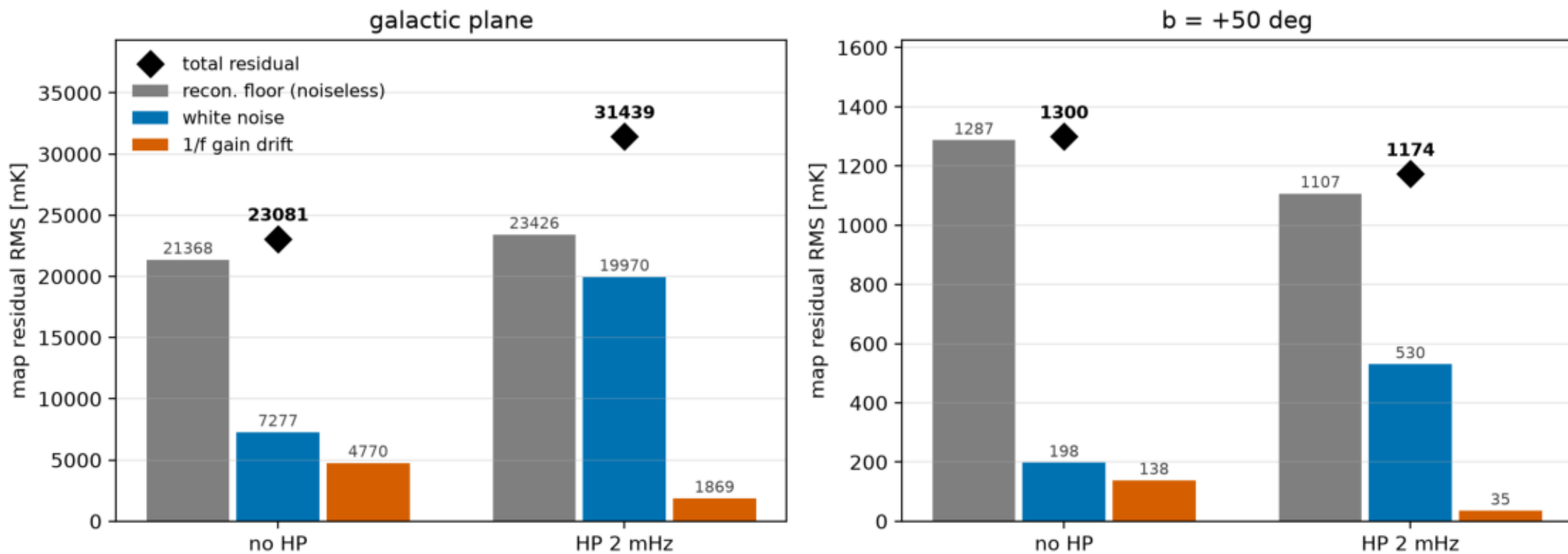


The high-pass only reaches the shaded band — and $1/f$ is already sub-dominant above it



galplane_1f_budget.png

Map-space noise budget: components add in quadrature to the total



galplane_1f_cutoff_sweep.png

Cutoff sweep — in the plane every cutoff is worse than no filter

